Applying Hölder's Inequality

Math 104 · Spring 2026

Statement

Theorem (Hölder's Inequality)**.** *Let $p, q > 1$ with $\dfrac{1}{p} + \dfrac{1}{q} = 1$ (conjugate exponents). If $f, g : [a, b] \to \mathbb{R}$ are Riemann integrable, then*

$$\int_a^b |f(x)\, g(x)|\, dx \;\le\; \|f\|_p \, \|g\|_q \;:=\; \left(\int_a^b |f|^p\right)^{1/p} \left(\int_a^b |g|^q\right)^{1/q}.$$

The discrete version: for sequences a_k, b_k,

$$\sum_{k=1}^n |a_k b_k| \le \left(\sum_{k=1}^n |a_k|^p\right)^{1/p} \left(\sum_{k=1}^n |b_k|^q\right)^{1/q}.$$

Remark. The case $p = q = 2$ is the **Cauchy–Schwarz inequality**. Young's inequality $uv \le u^p/p + v^q/q$ underlies the proof: normalize $u = |f|/\|f\|_p$, $v = |g|/\|g\|_q$, apply Young pointwise, then integrate.

Application 1: Bounding an Integral of a Product

Setup. You want to show $\int_a^b |h(x)|\, dx < \infty$ or bound it, where $h(x) = f(x)g(x)$ and neither factor alone is obviously integrable.

Technique Split $h = f \cdot g$. Choose conjugate p, q so that $\int |f|^p$ and $\int |g|^q$ are both finite. Then $\int |h| \le \|f\|_p \, \|g\|_q$.

Example 1. Show that $\displaystyle\int_0^1 \sqrt{x}\, e^{-x}\, dx$ is finite and bounded.

Take $f(x) = \sqrt{x} = x^{1/2}$ and $g(x) = e^{-x}$, with $p = 4$ and $q = 4/3$ (so $1/4 + 3/4 = 1$).

$$\int_0^1 \sqrt{x}\, e^{-x}\, dx \le \left(\int_0^1 x^2\, dx\right)^{1/4} \left(\int_0^1 e^{-4x/3}\, dx\right)^{3/4}.$$

Both factors are finite: $\int_0^1 x^2\, dx = 1/3$ and $e^{-4x/3}$ is bounded on $[0, 1]$.

Example 2. Show that $\displaystyle\int_0^1 \frac{f(x)}{\sqrt{x}}\, dx$ is finite whenever $f \in L^2[0, 1]$ (i.e., $\int_0^1 f^2\, dx < \infty$).

Apply Cauchy–Schwarz ($p = q = 2$) with $f(x)$ and $g(x) = 1/\sqrt{x}$:

$$\int_0^1 \frac{|f(x)|}{\sqrt{x}}\, dx \le \left(\int_0^1 f(x)^2\, dx\right)^{1/2} \left(\int_0^1 \frac{1}{x}\, dx\right)^{1/2}.$$

Wait — $\int_0^1 x^{-1}\,dx$ diverges, so $p = q = 2$ fails here. Instead take $p = 4$, $q = 4/3$:

$$\int_0^1 \frac{|f(x)|}{\sqrt{x}}\,dx \le \left(\int_0^1 |f|^4\right)^{1/4} \left(\int_0^1 x^{-2/3}\,dx\right)^{3/4}.$$

Now $\int_0^1 x^{-2/3}\,dx = 3$ (since $-2/3 > -1$), so this works whenever $f \in L^4[0,1]$.

Key lesson: The choice of p and q matters. If one choice makes a factor diverge, try different exponents.

Application 2: The Trivial Split ($g \equiv 1$)

Setup. You want to bound $\|h\|_1 = \int |h|$ using only information about $\|h\|_p$ for some $p > 1$.

Technique Write $h = h \cdot 1$ and apply Hölder with $f = h$, $g \equiv 1$:

$$\int_a^b |h|\,dx \le \left(\int_a^b |h|^p\right)^{1/p} (b-a)^{1/q}.$$

This converts an L^p bound into an L^1 bound, at the cost of a factor of $(b-a)^{1/q}$.

Example 3. Let f be continuous on $[0,1]$ with $\left(\int_0^1 f^2\right)^{1/2} \le 5$. Bound $\int_0^1 |f|$.

Apply Cauchy–Schwarz with $g \equiv 1$:

$$\int_0^1 |f|\,dx \le \left(\int_0^1 f^2\right)^{1/2} \left(\int_0^1 1^2\right)^{1/2} = 5 \cdot 1 = 5.$$

Example 4 (Comparing L^p norms on finite intervals)**.** For $1 \le r < s$ and $f \in L^s[a,b]$, show $\|f\|_r \le (b-a)^{1/r-1/s}\,\|f\|_s$.

Apply Hölder with $p = s/r > 1$ and $q = s/(s-r)$, writing $|f|^r = |f|^r \cdot 1$:

$$\int_a^b |f|^r \le \left(\int_a^b |f|^s\right)^{r/s} (b-a)^{1-r/s}.$$

Taking r-th roots: $\|f\|_r \le \|f\|_s \cdot (b-a)^{1/r-1/s}$.

Moral: On a finite interval, higher L^p norms dominate lower ones (up to a constant depending on interval length).

Application 3: Cauchy–Schwarz ($p = q = 2$)

Setup. The most common special case. Use when both factors are naturally square-integrable or when symmetry suggests equal exponents.

Technique $p = q = 2$: $\displaystyle\int |fg| \le \left(\int f^2\right)^{1/2} \left(\int g^2\right)^{1/2}$.

In $\mathbb{R}^n$ (discrete): $|\sum a_k b_k| \le \left(\sum a_k^2\right)^{1/2} \left(\sum b_k^2\right)^{1/2}$.

Example 5. Prove that $\left(\int_a^b f(x)\,dx\right)^2 \le (b-a)\int_a^b f(x)^2\,dx$.

This is the trivial split with Cauchy–Schwarz:

$$\left(\int_a^b f\right)^2 = \left(\int_a^b f\cdot 1\right)^2 \le \left(\int_a^b f^2\right)\left(\int_a^b 1\right) = (b-a)\int_a^b f^2.$$

Example 6. Prove the Cauchy–Schwarz inequality for inner products in $\mathbb{R}^n$: $|\langle u,v\rangle| \le \|u\|\|v\|$.

Apply the discrete Hölder with $p=q=2$ to sequences $a_k = u_k$, $b_k = v_k$:

$$\left|\sum_{k=1}^n u_k v_k\right| \le \sum_{k=1}^n |u_k||v_k| \le \left(\sum u_k^2\right)^{1/2}\left(\sum v_k^2\right)^{1/2} = \|u\|\|v\|.$$

Application 4: Bounding Improper Integrals (Comparison via Hölder)

Setup. You have an improper integral $\int_1^\infty f(x)/x^s\,dx$ and need to show it converges. If f is bounded, comparison suffices; but if f grows, Hölder can isolate the growth into a factor you can control.

Technique Write $f(x)/x^s = f(x)\cdot x^{-s}$. Choose p,q so that $\int |f|^p$ converges and $\int x^{-sq}$ converges (requires $sq>1$, i.e., $s>1/q = 1-1/p$).

Example 7. Show $\displaystyle\int_1^\infty \frac{x-[x]}{x^{s+1}}\,dx$ converges for all $s>0$.

Since $0 \le x-[x] < 1$, we have $\left|\dfrac{x-[x]}{x^{s+1}}\right| \le x^{-(s+1)}$. For $s>0$, $s+1>1$, so by comparison:

$$\int_1^\infty \frac{x-[x]}{x^{s+1}}\,dx \le \int_1^\infty x^{-(s+1)}\,dx = \frac{1}{s} < \infty.$$

(Here comparison suffices since the numerator is bounded. Hölder would be needed if f grew.)

Example 8. Show $\displaystyle\int_1^\infty \frac{\ln x}{x^s}\,dx$ converges for $s>1$.

Apply Hölder with $p=2$, $q=2$, writing $\ln x/x^s = (\ln x/x^{s/2})\cdot(1/x^{s/2})$:

$$\int_1^\infty \frac{\ln x}{x^s}\,dx \le \left(\int_1^\infty \frac{(\ln x)^2}{x^s}\,dx\right)^{1/2}\left(\int_1^\infty \frac{1}{x^s}\,dx\right)^{1/2}.$$

Both factors converge for $s>1$ (the second is $(s-1)^{-1/2}$; the first by integration by parts).

Application 5: Discrete Hölder for Finite Sums

Setup. You need to bound $\sum a_k b_k$ or $\sum |a_k|$ over a finite index set.

Technique Apply the discrete version. The key choice: if you know ℓ^p information about one sequence (e.g., it comes from a norm), match it with the conjugate exponent on the other.

Example 9. Prove that $\sum_{k=1}^{n} k \le n^{3/2} \left(\sum_{k=1}^{n} k^2\right)^{1/2} / n^{1/2}$ — actually, directly: $\sum_{k=1}^{n} k \le \sqrt{n} \left(\sum_{k=1}^{n} k^2\right)^{1/2}$.

Apply Cauchy–Schwarz with $a_k = 1$, $b_k = k$:

$$\sum_{k=1}^{n} k = \sum_{k=1}^{n} 1 \cdot k \le \left(\sum_{k=1}^{n} 1^2\right)^{1/2} \left(\sum_{k=1}^{n} k^2\right)^{1/2} = \sqrt{n} \left(\frac{n(n+1)(2n+1)}{6}\right)^{1/2}.$$

Example 10. Show that for $x_1, \ldots, x_n \ge 0$: $\left(\frac{x_1 + \cdots + x_n}{n}\right)^p \le \frac{x_1^p + \cdots + x_n^p}{n}$ for $p \ge 1$ (Jensen / power mean inequality).

By Hölder with $a_k = x_k$ and $b_k = 1/n$, exponents p and $q = p/(p-1)$:

$$\frac{\sum x_k}{n} = \sum x_k \cdot \frac{1}{n} \le \left(\sum x_k^p\right)^{1/p} \left(\sum \frac{1}{n^q}\right)^{1/q} = \left(\sum x_k^p\right)^{1/p} \cdot n^{1/q-1}.$$

Since $1/q - 1 = -1/p$: $(\sum x_k/n)^p \le \sum x_k^p/n$.

Application 6: Moment Generating Functions (Probability)

Setup. The **moment generating function** (MGF) of a random variable X is

$$M_X(t) = E[e^{tX}] = \int e^{tx}\, dP,$$

where dP is the probability measure. This is just a Lebesgue integral against a probability measure, so Hölder applies directly with $d\mu = dP$.

Log-Convexity of the MGF

Theorem 1. *$M_X(t)$ is **log-convex**: $\log M_X$ is a convex function of t.*

Proof. Let $\alpha \in (0,1)$, and set $p = 1/\alpha$, $q = 1/(1-\alpha)$, so $1/p + 1/q = 1$. Then:

$$M_X(\alpha s + (1-\alpha)t) = E\left[e^{\alpha s X} \cdot e^{(1-\alpha)tX}\right] \le \left(E[e^{sX}]\right)^{\alpha} \left(E[e^{tX}]\right)^{1-\alpha} = M_X(s)^{\alpha}\, M_X(t)^{1-\alpha},$$

where the inequality is Hölder with exponents $p = 1/\alpha$ and $q = 1/(1-\alpha)$. Taking log of both sides gives

$$\log M_X(\alpha s + (1-\alpha)t) \le \alpha \log M_X(s) + (1-\alpha) \log M_X(t),$$

which is exactly convexity of $\log M_X$. □

Technique To prove log-convexity of any quantity of the form $F(t) = E[f(t,X)]$ where $f(t,x) = e^{tx}$ or similar: write $F(\alpha s + (1-\alpha)t) = E[f(s,X)^{\alpha} f(t,X)^{1-\alpha}]$ and apply Hölder with $p = 1/\alpha$.

Remark. The **cumulant generating function** (CGF) is $K(t) = \log M_X(t)$. Its first derivative at 0 is $E[X]$; its second derivative at 0 is $\mathrm{Var}(X)$. Convexity of K means $K''(t) \ge 0$ everywhere, i.e., the "running variance" $\mathrm{Var}(e^{tX})$ is always nonneg — a tautology, but Hölder is what proves it globally, not just at $t = 0$.

Chernoff Bounds

Log-convexity is the engine behind the sharpest tail bounds in probability.

Example 11 (Chernoff bound)**.** For any $a \in \mathbb{R}$ and $t > 0$:

$$P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}} = e^{-ta} M_X(t).$$

Optimizing over $t > 0$ gives the **Chernoff bound**:

$$P(X \geq a) \leq \inf_{t>0} e^{-ta} M_X(t) = e^{-\sup_{t>0}(ta - \log M_X(t))}.$$

The exponent is the **Legendre transform** of $K(t) = \log M_X(t)$. Convexity of K (proved via Hölder above) guarantees the supremum is attained at a unique t^* where $K'(t^*) = a$, making the bound explicit and tight.

Example 12 (Large deviations / Cramér's theorem)**.** Let $S_n = (X_1 + \cdots + X_n)/n$ be the sample mean. The **rate function** governing how fast $P(S_n \geq a)$ decays is

$$I(a) = \sup_{t \in \mathbb{R}} (ta - K(t)),$$

the Legendre–Fenchel transform of K. Since K is convex (Hölder), I is a proper convex function and equals zero only at $a = E[X]$. Cramér's theorem says $P(S_n \geq a) \approx e^{-nI(a)}$ — so Hölder, via log-convexity, is literally the foundation of the exponential decay rate in the law of large numbers.

The Abstract Picture: Convex Duality

Both Hölder's inequality and MGFs are special cases of the same underlying structure in convex analysis.

Young's inequality — the engine of Hölder's proof — states:

$$uv \leq \frac{u^p}{p} + \frac{v^q}{q} \qquad \left(\tfrac{1}{p} + \tfrac{1}{q} = 1\right).$$

This is the **Fenchel–Young inequality** $\langle x, y \rangle \leq f(x) + f^*(y)$ applied to $f(u) = u^p/p$, whose convex conjugate is $f^*(v) = v^q/q$.

The Chernoff bound's optimization $\sup_t(ta - K(t))$ is exactly the convex conjugate $K^*(a)$ of the CGF. So the full chain is:

Young's inequality	$\longrightarrow$	Hölder's inequality
(Fenchel–Young, $f = u^p/p$)		(bound $E[\lvert fg \rvert]$)
Hölder applied to $E[e^{tX}]$	$\longrightarrow$	Log-convexity of $M_X(t)$
Legendre transform of $\log M_X$	$\longrightarrow$	Chernoff / Cramér rate function

Remark. At the deepest level, Hölder's inequality, MGFs, and large deviations are all facets of **convex duality**: the pairing between a convex function and its conjugate. Young's inequality is where it starts; the Cramér rate function is where it ends up.

Decision Guide: Choosing p and q

When you see $\int |fg|$ or $\sum |a_k b_k|$, ask:

1. **Is the product symmetric?** Use $p = q = 2$ (Cauchy–Schwarz). Fastest and most natural.
2. **Is one factor bounded?** Use comparison directly (Hölder is overkill). E.g., $|f| \leq M$ gives $\int |fg| \leq M \int |g|$.
3. **Does one factor have a singularity like $x^{-\alpha}$?** Choose q so that $q\alpha < 1$ (keeping the singular factor integrable). Then set $p = q/(q-1)$.
4. **Do you only know an L^p norm?** Use the trivial split with $g \equiv 1$ to get an L^1 bound. Result: $\int |h| \leq \|h\|_p (b-a)^{1/q}$.
5. **Does a weight function appear?** Write $\int w(x)|f(x)|\,dx = \int |f| \cdot |w|$ and put the weight w in the L^q factor (often easier to compute) and f in the L^p factor (often the unknown).

Remark. If your choice of p, q makes one factor infinite, try different exponents or split the integral into $\int_0^1 + \int_1^\infty$ and use different exponents on each piece.